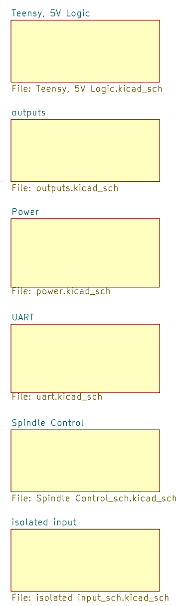


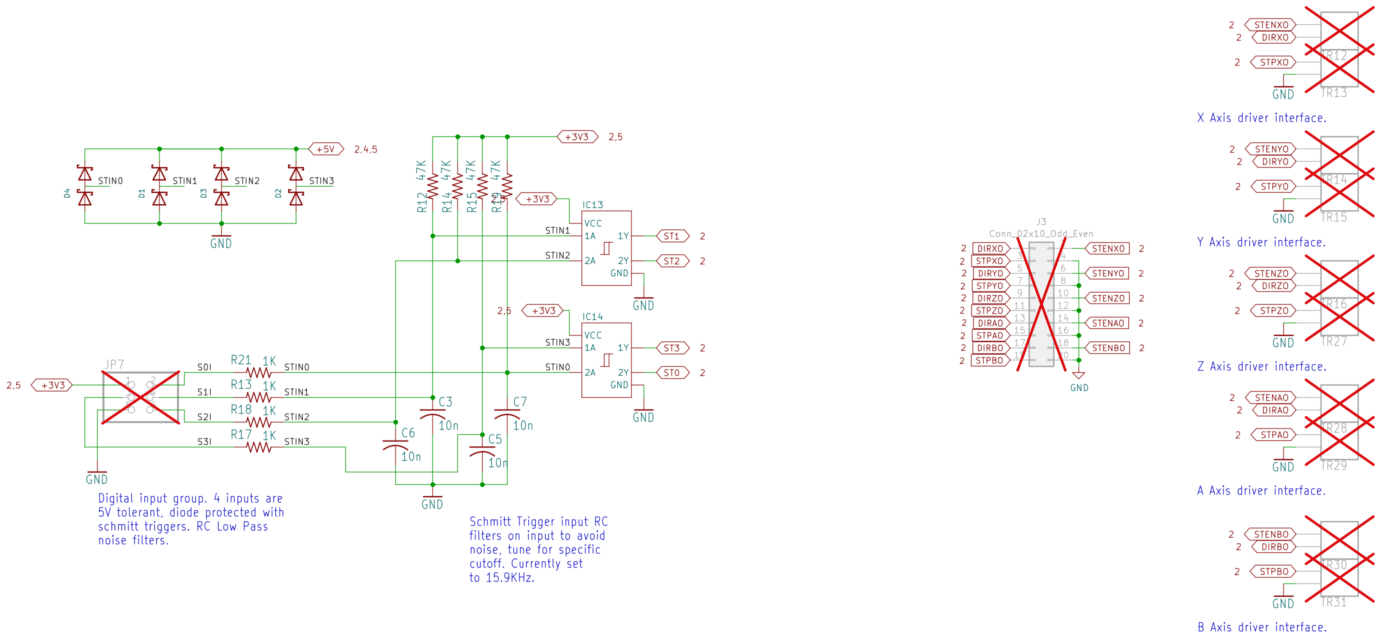
Brookwood Design

Sheet: /

File: V302.kicad_sch

Title: T41U5XBB

Size: User	Date: 2026-08-23	**Rev: V3.02**
KiCad E.D.A. 8.0.9		Id: 1/7



Digital input group. 4 inputs are 5V tolerant, diode protected with schmitt triggers. RC Low Pass noise filters.

Schmitt Trigger input RC filters on input to avoid noise, tune for specific cutoff. Currently set to 15.9KHz.

X Axis driver interface.

Y Axis driver interface.

Z Axis driver interface.

A Axis driver interface.

B Axis driver interface.

Brookwood Design

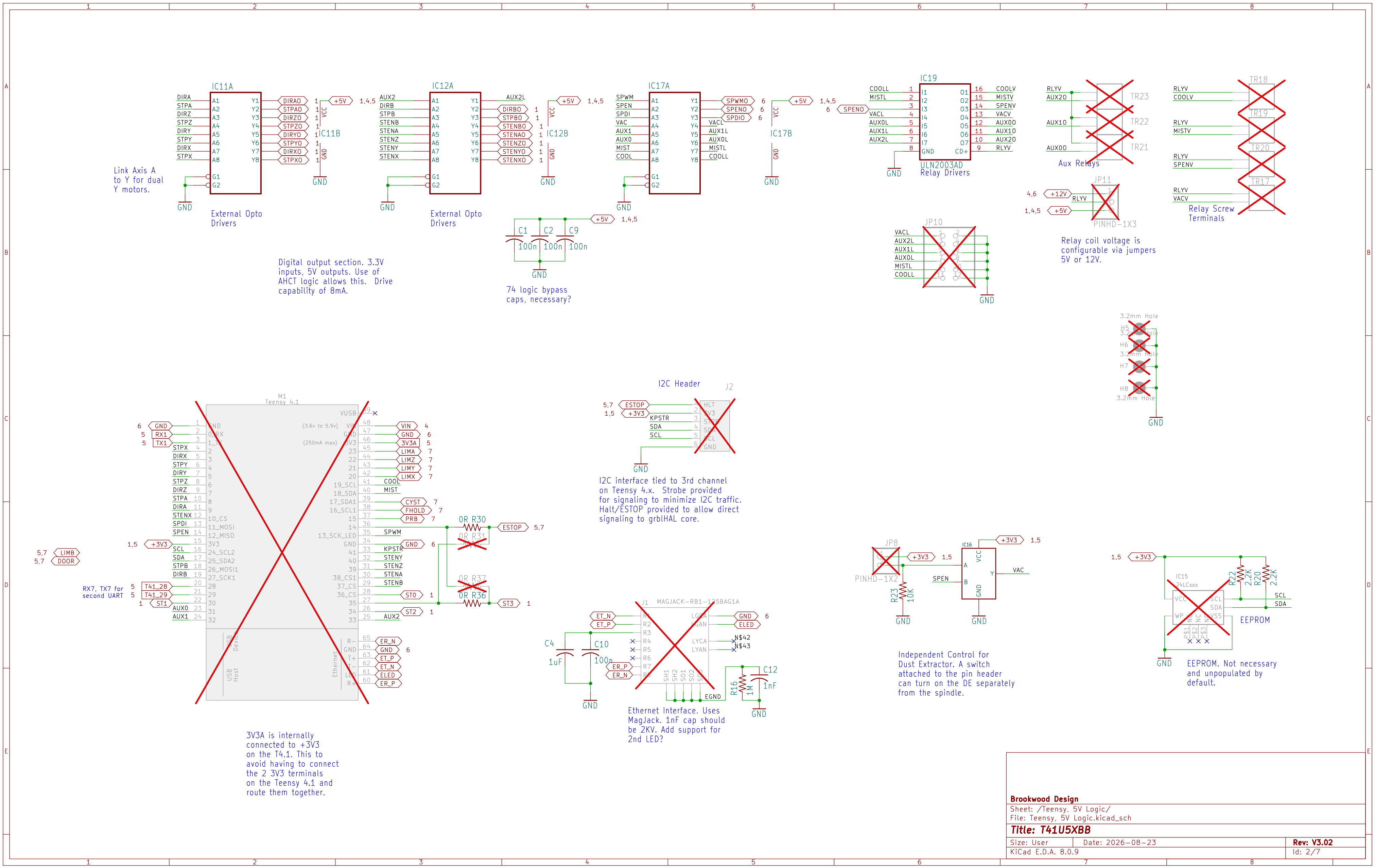
Sheet: /outputs/

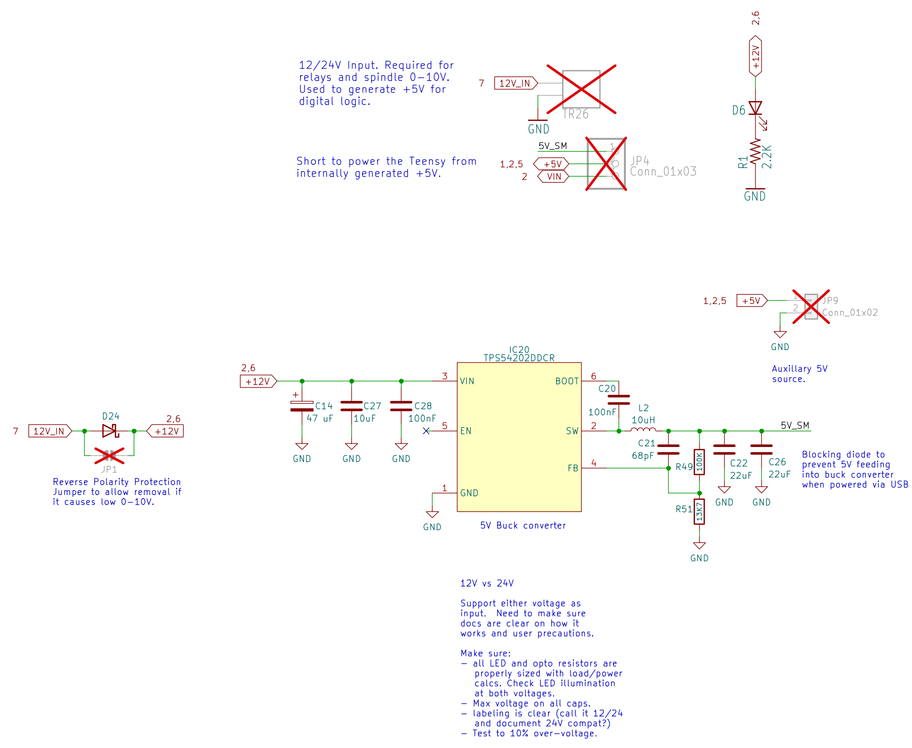
File: outputs.kicad_sch

Title: T41U5XBB

Size: User | Date: 2026-08-23 | **Rev: V3.02**

KiCad E.D.A. 8.0.9 | Id: 1/7


Link Axis A
to Y for dual
Y motors.
External Opto
Drivers
External Opto
Drivers
Digital output section. 3.3V
inputs, 5V outputs. Use of
AHCT logic allows this. Drive
capability of 8mA.
74 logic bypass
caps, necessary?
ULN2003AD
Relay Drivers
Aux Relays
Relay Screw
Terminals
Relay coil voltage is
configurable via jumpers
5V or 12V.
I2C Header
I2C interface tied to 3rd channel
on Teensy 4.x. Strobe provided
for signaling to minimize I2C traffic.
Halt/ESTOP provided to allow direct
signaling to grblHAL core.
RX7, TX7 for
second UART
3V3A is internally
connected to +3V3
on the T4.1. This to
avoid having to connect
the 2 3V3 terminals
on the Teensy 4.1 and
route them together.
Ethernet Interface. Uses
MagJack. 1nF cap should
be 2KV. Add support for
2nd LED?
Independent Control for
Dust Extractor. A switch
attached to the pin header
can turn on the DE separately
from the spindle.
EEPROM
EEPROM. Not necessary
and unpopulated by
default.
Brookwood Design
Sheet: /Teensy. 5V Logic/
File: Teensy. 5V Logic.kicad_sch
Title: T41U5XBB
Size: User
Date: 2026-08-23
Rev: V3.02
KiCad E.D.A. 8.0.9
Id: 2/7



Brookwood Design

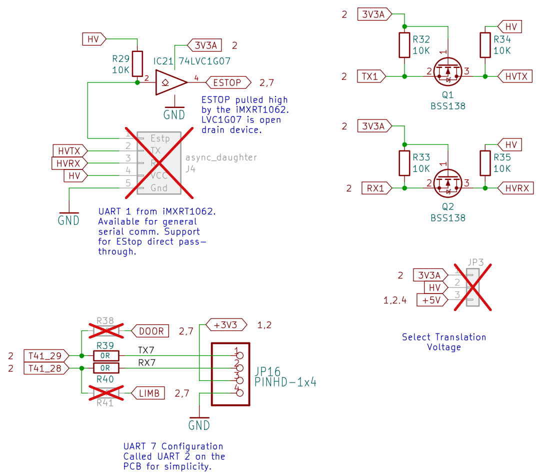
Sheet: /Power/

File: power.kicad_sch

Title: T41U5XBB

Size: User | Date: 2026-08-23 | **Rev: V3.02**

KiCad E.D.A. 8.0.9 | Id: 4/7



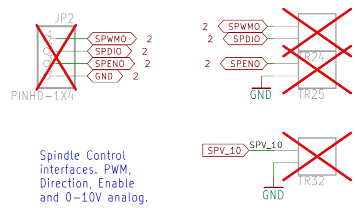
Brookwood Design

Sheet: /UART/

File: uart.kicad_sch

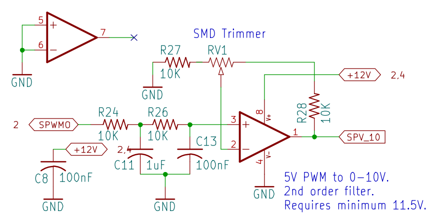
Title: T41U5XBB

Size: User	Date: 2026-08-23	**Rev: V3.02**
KiCad E.D.A. 8.0.9		Id: 5/7



Spindle Control
interfaces. PWM,
Direction, Enable
and 0-10V analog.

Spindle Control. Supports direct PWM and on/off (via Spindle enable) as well as 0-10V control. 10V output is adjustable via trim pot. Consider replacing LM358 with smaller opamp, single channel (JLC LM358 is Basic part and lowest cost Op Amp).

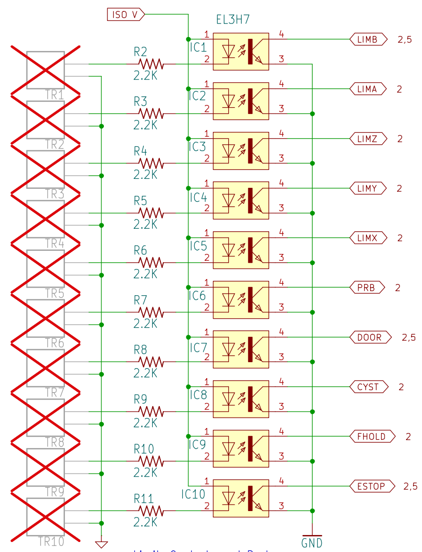


Brookwood Design

Sheet: /Spindle Control/
File: Spindle Control_sch.kicad_sch

Title: T41U5XBB

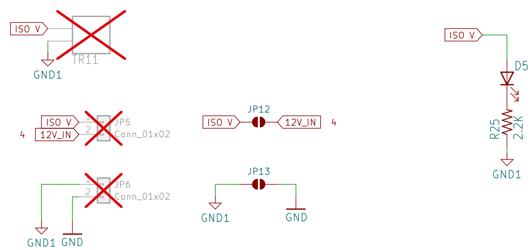
Size: User	Date: 2026-08-23	**Rev: V3.02**
KiCad E.D.A. 8.0.9		Id: 6/7



GND1 is the isolated ground plane.

Limit, Controls and Probe Interfaces.

12/24V operation on LED side. 3.3V operation on phototransistor side. LED If set to slightly higher than 10 mA.



Isolated Input side. Can be strapped to main power input or separately powered via isolated PSU

Brookwood Design
Sheet: /isolated input/
File: isolated input_sch.kicad_sch

Title: T41U5XBB

Size: User | Date: 2026-08-23 | **Rev: V3.02**
KiCad E.D.A. 8.0.9 | Id: 7/7